



Angiography-derived fractional flow reserve versus coronary angiography to guide coronary artery bypass grafting in patients undergoing surgical valve procedures with concomitant coronary artery disease in China (FAVOR IV-QVAS): a multicentre, triple-blind, randomised trial

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Summary

Background For patients undergoing surgical valve procedures with concomitant coronary artery disease, current guidelines recommend that coronary artery bypass grafting (CABG) should be anatomically guided on the basis of stenosis severity, as assessed by coronary angiography. We aimed to test whether a physiologically guided strategy using angiography-derived fractional flow reserve (FFR) could improve clinical outcomes in this population.

Methods FAVOR IV-QVAS is an investigator-initiated, multicentre, randomised, triple-blind trial done at 12 tertiary hospitals in China. Eligible patients were aged 18 years or older and were scheduled for valve surgery, with at least one clinically significant stenosis in a major coronary artery. Patients were randomly assigned (1:1) to undergo physiologically guided CABG (for lesions with an angiography-derived FFR value ≤ 0.80) or anatomically guided CABG (for lesions with a stenosis diameter $\geq 50\%$ on coronary angiography). Randomisation was done using a web-based program and stratified by site with fixed blocks of four. Patients, surgeons, follow-up physicians, and outcome assessors were masked to treatment allocation. The primary outcome was a composite of death, myocardial infarction, stroke, unplanned coronary revascularisation, and new renal failure requiring dialysis within 30 days after surgery. The key secondary outcome was a composite of death, myocardial infarction, stroke, unplanned coronary revascularisation, and hospitalisation for unstable angina or heart failure at a minimum follow-up of 1 year. The primary analysis of the primary and key secondary outcomes was done in a modified intention-to-treat population that included all randomly assigned patients who underwent surgery and had available data for the primary outcome. Missing data for the primary outcome were planned to be analysed using complete-case analysis or multiple imputation, with a proportion of missing data of 2% as the threshold. This trial is registered at ClinicalTrials.gov (NCT03977129); extended follow-up is ongoing.

Findings Between Aug 4, 2019, and Aug 13, 2024, 793 patients were enrolled. 396 were randomly assigned to the angiography-derived FFR group and 397 to the coronary angiography group; one patient in the coronary angiography group declined surgery and was excluded from the modified intention-to-treat population. The median age was 65 years (IQR 59–70), 221 (28%) patients were female, and 571 (72%) were male. Concomitant CABG was done in 223 (56%) patients in the angiography-derived FFR group and in 388 (98%) patients in the coronary angiography group. The primary outcome occurred in 31 (7.8%) patients in the angiography-derived FFR group and 53 (13.4%) in the coronary angiography group (absolute difference -5.6 percentage points [95% CI -9.9 to -1.3]; risk ratio 0.58 [95% CI 0.38 to 0.89]; $p=0.011$). Death within 30 days occurred in 11 (2.8%) patients in the angiography-derived FFR group and 17 (4.3%) patients in the coronary angiography group. At a median follow-up of 27 months (28 months [IQR 18–44] in the angiography-derived FFR group and 27 months [18–42] in the coronary angiography group), the key secondary outcome occurred in 82 (20.7%) patients in the angiography-derived FFR group and in 106 (26.8%) patients in the coronary angiography group (hazard ratio 0.74 [95% CI 0.55–0.98]; $p=0.036$).

Interpretation Among patients undergoing valve surgery with concomitant coronary artery disease, physiologically guided CABG using angiography-derived FFR reduced the incidence of the composite perioperative outcome compared with anatomically guided CABG. These findings support a selective approach to surgical coronary revascularisation guided by physiological assessment in patients undergoing valve procedures.

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See Online for appendix 1

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Introduction

Coronary artery disease is common in patients undergoing surgical valve procedures, with a reported prevalence of approximately 20%, depending on definitions and type of valve disease.¹ Current guidelines recommend that concomitant coronary artery bypass grafting (CABG) should be guided anatomically, on the basis of stenosis severity on coronary angiography, although high-quality evidence from randomised trials supporting this approach is scarce.^{2,3} The perioperative morbidity and mortality associated with combined valve and CABG procedures are higher than for either operation alone.^{4–6}

Among patients undergoing percutaneous coronary interventions, physiological rather than anatomical guidance has been shown to improve clinical outcomes^{7,8} partly by avoiding the procedural risk associated with treating non-ischaemic lesions. Data on the use of physiological assessment to guide CABG are scant,^{9,10} and only a small number of studies have been done in patients with concomitant valvular and coronary disease.^{11,12}

Angiography-derived fractional flow reserve (FFR) is a computational method that assesses coronary stenosis without a pressure wire or hyperaemia, and has been shown to correlate closely with wire-based FFR.^{13,14} Angiography-derived FFR guidance has been tested for

percutaneous coronary interventions,^{15–17} but has not been evaluated in randomised trials for CABG. Unlike wire-based FFR, angiography-derived FFR is less affected by aortic stenosis because it relies on resting contrast flow rather than hyperaemia.¹⁸

This trial aimed to test the hypothesis that, in patients undergoing valve surgery with concomitant coronary artery disease, angiography-derived FFR-guided (physiologically guided) CABG would improve postoperative clinical outcomes compared with coronary angiography-guided (anatomically guided) CABG.

Methods

Study design and participants

The Quantitative Flow Ratio Guided Coronary Artery Bypass Grafting for Patients Undergoing Primary Valve Surgery with Concomitant Coronary Artery Disease (FAVOR IV-QVAS) trial was an investigator-initiated, multicentre, triple-blind, randomised clinical trial conducted at 12 tertiary hospitals in China. The protocol was first approved by the institutional review board of Ruijin Hospital Shanghai Jiao Tong University School of Medicine (KJ-2019–66) and then approved by the institutional review board at each participating site. The trial protocol and statistical analysis plan are provided in appendix 1.

Research in context

Evidence before this study

For patients undergoing valve surgery with concomitant coronary artery disease, current guidelines recommend that coronary artery bypass grafting (CABG) be anatomically guided on the basis of stenosis severity as assessed by coronary angiography. In percutaneous coronary intervention, physiologically guided revascularisation improves outcomes by avoiding treatment of non-ischaemic lesions, but its role in CABG remains uncertain, and few previous studies have addressed this question. We searched PubMed for randomised controlled trials and observational studies published from database inception to Oct 29, 2025, without language restrictions, using the search terms “fractional flow reserve” OR “FFR” OR “quantitative flow ratio” OR “QFR” OR “angiography-derived FFR” OR (“angioFFR” AND “coronary artery bypass grafting”) OR (“CABG” AND “valve surgery”). Our search identified only small observational studies that assessed wire-based or angiography-derived indices to guide CABG, and none provided adequately powered evidence in patients undergoing valve surgery with concomitant coronary disease.

Added value of this study

To the best of our knowledge, FAVOR IV-QVAS is the first large, multicentre, triple-blind, randomised trial of

physiologically guided CABG using angiography-derived fractional flow reserve (FFR) in patients undergoing valve surgery. In this trial, conducted at 12 Chinese sites and enrolling 793 patients, angiography-derived FFR guidance significantly reduced the mean number of grafts placed per patient, shortened bypass times, and reduced the composite of death, myocardial infarction, stroke, unplanned revascularisation or new dialysis at 30 days, with benefits maintained at a median follow-up of 27 months.

Implications of all the available evidence

These findings support a selective, physiologically guided approach to concomitant CABG for patients undergoing valve surgery. Physiological assessment can identify non-ischaemic lesions for which surgery can be deferred, simplifying surgery and reducing perioperative morbidity. Further research is needed to ascertain the long-term durability of this approach, its applicability to broader patient populations and health-care systems, and its impact on health-related quality of life and cost-effectiveness.

A steering committee including a patient representative designed the protocol and oversaw the trial. The study was conducted in accordance with all applicable clinical trial regulations and guidelines. The regulatory sponsor was Ruijin Hospital Shanghai Jiao Tong University School of Medicine (Shanghai, China). The collaborator was the Med-X Research Institute, Shanghai Jiao Tong University. The trial was monitored by Caidya Biopharmaceutical and coordinated by CCBR Clinical Research (Beijing), SMO Clinplus, and Beijing Shiyan Pharmaceutical Technology. Data management and statistical analyses were done by the Department of Biostatistics, School of Public Health, Fudan University (Shanghai, China). An independent clinical outcome committee, comprising two cardiac surgeons and three cardiologists who were not involved in the trial, adjudicated all clinical outcomes. Given that the tested interventions are routine practice, the Ruijin Hospital Shanghai Jiao Tong University School of Medicine ethics committee did not require formal oversight by a data safety monitoring board. Instead, all serious adverse events were reported to the ethics committee of each participating site within 7 days of occurrence. Trial study sites, investigators, collaborators, and organisations are listed in appendix 1 (pp 4–6). This trial was registered with ClinicalTrials.gov (NCT03977129); long-term follow-up is ongoing.

Eligible patients were adults aged 18 years or older who were scheduled to undergo elective valve surgery for primary aortic valve disease, mitral valve disease, or both, and who had at least one major coronary artery with angiographic diameter stenosis of 50% or greater on coronary angiography, in a vessel of diameter 1.5 mm or greater and suitable for CABG. Key exclusion criteria included previous cardiac surgery, planned staged percutaneous coronary intervention, secondary valve disease, cardiogenic shock, poor angiographic image quality precluding analysis by angiography-derived FFR, or life expectancy of less than 3 years. The full list of inclusion and exclusion criteria is provided in appendix 1 (pp 48–49). Patients provided written informed consent before being enrolled into the trial. Patient sex was self-reported as male or female, and race or ethnicity data were self-reported as East Asian or other.

Randomisation and masking

An independent panel of surgeons developed two separate CABG plans, specifying target vessels, for each eligible patient: one based on an angiography-derived FFR-guided strategy and the other based on a coronary angiography-guided strategy. The CABG plans were prepared in advance, masked, and labelled as plan A or plan B before randomisation, with no consistent correspondence between plan A and plan B or angiography-derived FFR-guided and coronary angiography-guided to maintain masking.

An independent statistician generated the randomisation sequence using SAS software (version 9.4),

stratified by site with fixed blocks of four, and implemented through a secure web-based system. Randomisation was then done by an unmasked staff member at a core laboratory. Eligible patients were randomly assigned in a 1:1 ratio to either angiography-derived FFR-guided CABG or coronary angiography-guided CABG during valve surgery. The operating surgeon was informed only of the allocated plan (A or B) to be followed, not the underlying guidance strategy. Patients, follow-up physicians, and outcome assessors were also masked to the treatment assignments.

Procedures

Before randomisation, coronary angiograms were transmitted to an independent core laboratory (Shanghai Jiao Tong University–Pulse Medical Imaging joint laboratory) for analysis using angiography-derived FFR. This was done for all eligible consenting patients. Angiography-derived FFR was assessed using either quantitative flow ratio (QFR)¹⁹ or the upgraded version Murray law-based version of QFR (μ QFR) on the AngioPlus system (Pulse Medical, Shanghai, China)²⁰ by certified analysts. In μ QFR, an artificial intelligence algorithm is used for automatic delineation of coronary vessels from a single angiographic view, and Murray's bifurcation fractal law is incorporated to compute FFR in both the main and side branches. Standard operating procedures for angiography-derived FFR are provided in appendix 1 (pp 8–13).

In patients randomly assigned to the plan derived from the angiography-derived FFR-guided strategy, concomitant CABG would be done only if the angiography-derived FFR value of the target vessel was no greater than 0.80. In patients randomly assigned to the plan derived from the coronary angiography-guided strategy, concomitant CABG would be done for all vessels with coronary angiography-estimated diameter stenosis of at least 50%. Surgical procedures were then done according to the informed CABG plan with cardiopulmonary bypass and cardioplegic arrest in the standard fashion. Postoperative medical therapy, including antithrombotic and other secondary prevention medications, was administered according to current guidelines.

Patients had an in-hospital follow-up visit at 1 week post-surgery (before discharge). Subsequently, patients had outpatient visits at 1, 6, and 12 months postoperatively, then every 6 months by telephone or visit by local physician. During outpatient visits, patients underwent blood tests, electrocardiograms, echocardiograms, and coronary CT angiograms at 1 and 3 years. Adverse events were assessed at the designated follow-up visits, with severity graded as mild, moderate, or severe, and treatment relatedness adjudicated by the follow-up physicians.

Outcomes

The primary outcome was a composite of all-cause death, myocardial infarction, stroke, unplanned coronary

revascularisation, and new renal failure requiring dialysis within 30 days after surgery. The key secondary outcome was a composite of all-cause death, myocardial infarction, stroke, unplanned coronary revascularisation, hospitalisation for unstable angina, and hospitalisation for heart failure at 1 year and 3 years of follow-up. Other secondary outcomes were graft failure (stenosis $\geq 50\%$ or occlusion) at 1 and 3 years; health-related quality-of-life (EQ-5D) score at 30 days, 1 year, and 3 years; and cost-effectiveness analyses. Prespecified exploratory outcomes included the number of bypass grafts per patient, cardiopulmonary bypass time, aortic cross-clamp time, red blood cell transfusion, hospital stay after surgery, change of angina pectoris assessed by Canadian Cardiovascular Society grades, and heart function assessed by New York Heart Association classes between baseline and 1 year, and between baseline and 3 years of follow-up after surgery. Safety outcomes comprised

adverse events of interest (major surgical complications) within 30 days and other serious adverse events during follow-up. The definition of clinical outcomes is listed in appendix 1 (pp 14–18).

The primary outcome at 30 days, key secondary outcomes at 1 year and 3 years of follow-up, and safety outcomes are reported in the current paper. Analyses of graft outcomes, health-related quality of life, and cost-effectiveness are ongoing and will be reported in subsequent publications after ongoing extended follow-up.

Statistical analysis

The sample size calculation was based on the assumption that the 30-day incidence of the composite outcome would be 12% in the coronary angiography group and 6% in the angiography-derived FFR group (risk ratio [RR] 0·5). The incidences were estimated from the 2015 Society of Thoracic Surgeons database and our pilot study.^{11,21} With a two-sided type I error rate of 0·05 and an anticipated 10% dropout rate, a total of 396 patients per group (792 in total) was required to achieve 80% power to show the superiority of the angiography-derived FFR group over the coronary angiography group with respect to the primary outcome.

The primary analysis was conducted in the modified intention-to-treat population, which included all randomised patients who underwent surgery and had available data for the primary outcome. Safety outcomes were assessed in the safety population, consisting of all randomised patients who underwent surgery and for whom safety outcome data were available. Sensitivity analyses were done in the per-protocol population, defined as the subset of the intention-to-treat population without major deviations from the eligibility criteria or the allocated CABG strategy. A hierarchical testing procedure was prespecified: the key secondary outcome was to be formally tested in the modified intention-to-treat population only if the between-group difference for the primary outcome reached statistical significance. If the null hypothesis for the primary outcome was not rejected, the key secondary outcome was not intended to support formal inferential claims.

Formal hypothesis testing of the primary outcome was done with the Pearson χ^2 test. 95% CIs for the incidence in each group were calculated with the Clopper–Pearson method, and the 95% CI for the absolute difference between two groups was estimated with the Newcombe–Wilson method. When the proportion of missing data for the primary outcome was 2% or less, complete-case analysis was planned as the primary approach, with multiple imputation performed for sensitivity analysis. When the proportion of missing data exceeded 2%, multiple imputation was prespecified as the primary method, and complete-case analysis was used for sensitivity analysis. No interim analysis was included in this trial.

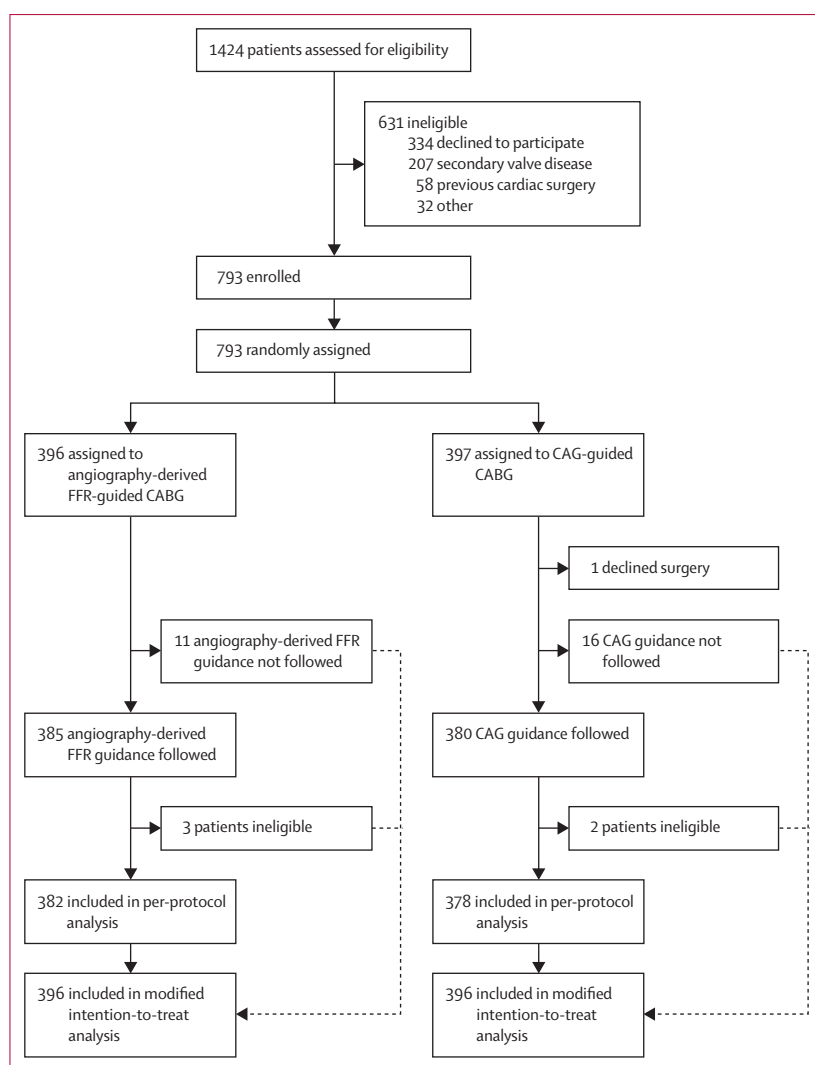


Figure 1: Trial profile

CABG=coronary artery bypass grafting. CAG=coronary angiography. FFR=fractional flow reserve.

Formal hypothesis testing of the key secondary outcomes in the hierarchical sequence was done with the log-rank test applied to Kaplan–Meier estimates of the cumulative incidence of events. Cox proportional hazards models were used to estimate hazard ratios (HRs) with 95% CIs. The proportional hazards assumption was evaluated by means of an extended Cox model that included a treatment-by-log (time) interaction term. The prespecified statistical analysis plan stipulated that, if the interaction was significant ($p < 0.05$), the primary analysis would instead use a logistic regression model with the natural logarithm of follow-up time as an offset, to provide a time-averaged treatment effect. For composite outcomes and their components, results are presented as overall event rates, landmark rates at 1 year and 3 years, and incidence densities (per 100 patient-years).

Further details regarding the analysis of other secondary outcomes, exploratory outcomes, safety outcomes, and subgroups are provided in the statistical analysis plan (appendix 1 pp 65–82). A two-sided $p < 0.05$ was considered statistically significant. All statistical analyses were done using SAS software, version 9.4.

Role of the funding source

The funders of the trial had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

Between Aug 4, 2019, and Aug 13, 2024, 1424 patients with planned valve surgery and concomitant coronary artery lesions with stenosis of at least 50% were screened for eligibility. 793 patients were enrolled (figure 1; appendix 1 p 7), of whom 396 were assigned to the angiography-derived FFR group and 397 to the coronary angiography group. All patients completed the 30-day follow-up; 792 were included in the modified intention-to-treat population analysis, and one patient in the coronary angiography group declined surgery. The per-protocol population comprised 760 patients (382 in the angiography-derived FFR group and 378 in the coronary angiography group). Five patients were excluded for ineligibility (three from the angiography-derived FFR group and two from the coronary angiography group) and 27 for surgeon non-adherence to the allocated CABG strategy (11 from the angiography-derived FFR group and 16 from the coronary angiography group). Protocol adherence was 97% (382 of 396) in the angiography-derived FFR group and 96% (378 of 396) in the coronary angiography group (appendix 1 p 22).

The baseline characteristics of patients are summarised in table 1. Median age of included patients was 65 years (IQR 59–70), 221 (28%) of 792 were female, and 571 (72%) of 792 were male. Degenerative disease was the most common cause of valvular dysfunction (586 [74%] patients), followed by rheumatic disease (174 [22%]). The most common valvular disease was mitral

	Angiography-derived FFR group (n=396)	Coronary angiography group (n=396)
Age, years	65 (59–70)	65 (59–70)
Sex		
Female	112 (28%)	109 (28%)
Male	284 (72%)	287 (72%)
Race or ethnicity		
East Asian	396 (100%)	396 (100%)
BMI, kg/m ²	24.6 (22.6–27.0)	24.2 (22.0–26.6)
Aetiology of valvular disease		
Degenerative	293 (74%)	293 (74%)
Rheumatic	93 (23%)	81 (20%)
Other	10 (3%)	22 (6%)
Types of valvular pathology		
Aortic stenosis	96 (24%)	90 (23%)
Aortic regurgitation	131 (33%)	158 (40%)
Mitral stenosis	60 (15%)	56 (14%)
Mitral regurgitation	206 (52%)	218 (55%)
Isolated aortic valve disease	146 (37%)	144 (36%)
Isolated mitral valve disease	192 (48%)	174 (44%)
Combined aortic and mitral valve disease	56 (14%)	77 (19%)
Tricuspid regurgitation	82 (21%)	86 (22%)
Previous myocardial infarction	16 (4%)	29 (7%)
Previous stroke	50 (13%)	53 (13%)
Hypertension	284 (72%)	274 (69%)
Diabetes	91 (23%)	106 (27%)
Atrial fibrillation	125 (32%)	113 (29%)
Current or former smoker	179 (45%)	181 (46%)
Left ventricular ejection fraction	61% (54–65)	61% (55–66)
SYNTAX score*	7 (5–12)	8 (5–13)
Coronary artery lesion		
Left main artery	10 (3%)	15 (4%)
Single vessel	230 (58%)	216 (55%)
Two vessels	95 (24%)	107 (27%)
Three vessels	70 (18%)	72 (18%)
STS-PROM score, %†	2.7% (1.7–4.3)	2.9% (1.9–4.8)

Data are n (%) or median (IQR). FFR=fractional flow reserve. STS-PROM=Society of Thoracic Surgeons predicted risk of mortality. SYNTAX=synergy between percutaneous coronary intervention with taxus and cardiac surgery.

*The SYNTAX score reflects a comprehensive angiographic assessment of the coronary vasculature; a higher score denotes higher anatomical complexity, and a score of 22 or less typically indicates low anatomical complexity of coronary artery disease. The score was calculated by the core laboratory. †The STS-PROM score reflects operative mortality defined as: (1) all-cause death during the hospital stay for the operation, even if after 30 days (including patients transferred to other acute care facilities); and (2) all-cause deaths after discharge from the hospital but before the end of the 30th postoperative day. The score was calculated as valve surgery plus coronary artery bypass grafting here.

Table 1: Patient characteristics at baseline (modified intention-to-treat population)

regurgitation (424 [54%] patients), followed by aortic regurgitation (289 [36%]), aortic stenosis (186 [23%]), and mitral stenosis (116 [15%]). 558 (70%) patients had hypertension, 197 (25%) had diabetes, and 238 (30%) had preoperative atrial fibrillation. The median left ventricular ejection fraction was 61% (IQR 54–66), the median synergy between percutaneous coronary intervention with taxus and cardiac surgery (SYNTAX) score was 8 (IQR 5–12), and the median Society of Thoracic Surgeons

	Angiography-derived FFR group (n=396)	Coronary angiography group (n=396)	Absolute difference, percentage points (95% CI)	Risk ratio (95% CI)	p value
Primary composite outcome					
Death, myocardial infarction, stroke, unplanned revascularisation, or new renal failure requiring dialysis	31 (7.8%)	53 (13.4%)	-5.6 (-9.9 to -1.3)	0.58 (0.38-0.89)	0.011
Individual components of the primary outcome					
Death	11 (2.8%)	17 (4.3%)	-1.5 (-4.3 to 1.1)	0.65 (0.31-1.36)	..
Cardiovascular death	8 (2.0%)	12 (3.0%)	..	..	..
Non-cardiovascular death	3 (0.8%)	5 (1.3%)	..	..	..
Myocardial infarction	6 (1.5%)	22 (5.6%)	-4.0 (-6.9 to -1.5)	0.27 (0.11-0.67)	..
Stroke	6 (1.5%)	9 (2.3%)	-0.8 (-2.9 to 1.3)	0.67 (0.24-1.86)	..
Unplanned revascularisation	5 (1.3%)	1 (0.3%)	1.0 (-0.4 to 2.7)	5.00 (0.59-42.60)	..
New renal failure requiring dialysis	9 (2.3%)	18 (4.6%)	-2.3 (-5.0 to 0.3)	0.50 (0.23-1.10)	..

Data are n (%) unless otherwise specified. FFR=fractional flow reserve.

Table 2: Primary outcome at 30 days (modified intention-to-treat population)

predicted risk of mortality (STS-PROM) score was 2.8% (IQR 1.8–4.5).

In this trial, angiography-derived FFR was assessed using QFR in the first 117 (15%) patients and μ QFR in the remaining 675 (85%) patients. Characteristics of the target vessels are reported in appendix 1 (p 23). 1416 target vessels were included, with 699 (49%) in the angiography-derived FFR group and 717 (51%) in the coronary angiography group. The left anterior descending artery was the most frequently diseased vessel, found in 745 (53%) target vessels (374 [54%] in the angiography-derived FFR group and 371 [52%] in the coronary angiography group). At visual estimation, the median stenosis severity was 70% (IQR 60–80) in both the angiography-derived FFR group and the coronary angiography group, with most lesions (335 [48%] in the angiography-derived FFR group and 356 [50%] in the coronary angiography group) falling in the 70–89% stenosis range. At quantitative coronary angiography the median diameter stenosis was 47% (IQR 39–56) and the median area stenosis was 71% (IQR 62–78). Angiography-derived FFR assessment showed a mean value of 0.75 (SD 0.19) and median value of 0.78 (IQR 0.70–0.87), with 753 (53%) of vessels with an angiography-derived FFR value no greater than 0.80. Of note, in the angiography-derived FFR group, 189 (80%) of 235 vessels with visual 50–69% stenosis and 128 (36%) of 356 vessels with visual 70–89% stenosis had an angiography-derived FFR value greater than 0.80.

Characteristics of surgical procedures are reported in appendix 1 (p 24). Patients in the angiography-derived FFR group were less likely to receive CABG than those in the coronary angiography group (223 [56%] of 396 vs 388 [98%] of 396) and received fewer grafts per patient (mean 0.9 [SD 1.0] vs 1.8 [1.1], median 1.0 [IQR 0–1.0] vs 1.0 [1.0–2.0]). Patients in the angiography-derived FFR group also had shorter cardiopulmonary bypass times (median 127 min [IQR 95–170] vs 150 min [116–193]) and

aortic cross-clamp times (median 84 min [IQR 60–112] vs 93 min [67–124]), and fewer red blood cell transfusions (median 0 units [IQR 0–3] vs 2 units [0–4]) than those in the coronary angiography group. The median hospital stay post-surgery was 11 days (IQR 8–14) in the angiography-derived FFR group versus 12 days (9–16) in the coronary angiography group.

The primary outcome occurred in 31 of 396 patients in the angiography-derived FFR group and in 53 of 396 patients in the coronary angiography group within 30 days after surgery (7.8% vs 13.4%; absolute difference, -5.6 percentage points [95% CI -9.9 to -1.3]; RR 0.58 [95% CI 0.38 to 0.89]; $p=0.011$; table 2; appendix 1 p 19). The reduction was driven mainly by a lower incidence of myocardial infarction (six [1.5%] patients vs 22 [5.6%] patients; RR 0.27 [95% CI 0.11–0.67]; appendix 1 pp 34–35). Death occurred in 11 (2.8%) patients in the angiography-derived FFR group versus 17 (4.3%) in the coronary angiography group, stroke in six (1.5%) versus nine (2.3%), and new dialysis in nine (2.3%) versus 18 (4.6%). Of the 28 deaths, 20 (eight in the angiography-derived FFR group vs 12 in the coronary angiography group) were due to cardiovascular causes, and eight (three vs five) were due to non-cardiovascular causes. Unplanned revascularisation occurred in five [1%] patients in the angiography-derived FFR group versus one (<1%) in the coronary angiography group. Because all patients in the modified intention-to-treat population completed the 30-day visit, complete-case analysis was applied for the primary outcome. The results of the per-protocol analysis were consistent with the primary analysis (appendix 1 p 25).

For time-to-event analyses, minimum 1-year follow-up was completed for 786 (99%) of 792 patients: 393 (99%) in the angiography-derived FFR group and 393 (99%) in the coronary angiography group. Median follow-up was 27 months (28 months [IQR 18–44] in the

	Angiography-derived FFR group (n=396)				Coronary angiography group (n=396)			
	Number of patients with events (%)	Patients with events per 100 patient-years	Cumulative incidence at 1 year, % (95% CI)	Cumulative incidence at 3 years, % (95% CI)	Number of patients with events (%)	Patients with events per 100 patient-years	Cumulative incidence at 1 year, % (95% CI)	Cumulative incidence at 3 years, % (95% CI)
Key secondary composite outcome								
Death, myocardial infarction, stroke, unplanned revascularisation, hospitalisation for unstable angina, and hospitalisation for heart failure	82 (20.7%)	8.8	14.4% (10.9–17.8)	21.3% (16.7–25.7)	106 (26.8%)	12.1	19.0% (15.0–22.8)	29.3% (24.0–34.2)
Component outcomes								
Death	42 (10.6%)	4.2	7.4% (4.7–9.9)	11.8% (8.2–15.3)	57 (14.4%)	5.8	8.1% (5.4–10.8)	16.5% (12.0–20.7)
Myocardial infarction	13 (3.3%)	1.3	2.8% (1.6–5.0)	3.3% (1.8–5.8)	28 (7.1%)	3.0	6.3% (4.3–9.2)	7.5% (5.2–10.8)
Stroke	29 (7.3%)	3.0	4.8% (3.1–7.5)	6.9% (4.7–10.1)	19 (4.8%)	2.0	3.0% (1.7–5.3)	5.4% (3.5–8.5)
Unplanned revascularisation	8 (2.0%)	0.8	1.8% (0.8–3.7)	2.1% (1.0–4.1)	1 (0.3%)	0.1	0.3% (0–1.8)	0.3% (0–1.8)
Hospitalisation for unstable angina	1 (0.3%)	0.1	0.3% (0–1.8)	0.3% (0–1.8)	2 (0.5%)	0.2	0.5% (0.1–2.0)	0.5% (0.1–2.0)
Hospitalisation for heart failure	11 (2.8%)	1.1	1.3% (0.5–3.0)	3.1% (1.6–5.8)	27 (6.8%)	2.9	4.6% (2.9–7.2)	7.3% (4.9–11.1)

FFR=fractional flow reserve.

Table 3: Key secondary outcome at 1 year and 3 years (modified intention-to-treat population)

angiography-derived FFR group and 27 months [18–42] in the coronary angiography group). The key secondary outcome occurred in 82 (21%) of 396 patients in the angiography-derived FFR group and in 106 (27%) of 396 patients in the coronary angiography group (HR 0.74 [95% CI 0.55–0.98]; $p=0.036$; table 3; figure 2). There was no significant interaction between randomised treatment and log(time) in the extended Cox model ($p=0.18$).

Major surgical complications within 30 days occurred in 120 (30%) of 396 patients in the angiography-derived FFR group and in 139 (35%) of 396 patients in the coronary angiography group (appendix 1 p 32). New-onset atrial fibrillation or atrial flutter was the most common complication, reported in 60 (15%) patients versus 57 (14%) patients. Pericardial or pleural effusion was the most common serious adverse event, reported in 14 (4%) patients versus 12 (3%) patients. Other serious adverse events during follow-up are summarised in appendix 1 (p 33).

The prespecified subgroup analyses of the primary outcome are shown in figure 3 and the post-hoc site effect analysis is shown in appendix 1 (p 26). There was no evidence of heterogeneity of the treatment effect across prespecified subgroups or across sites (all interaction p values >0.05).

Results for the secondary outcome were consistent in the per-protocol population (appendix 1 p 20). A post-hoc landmark analysis showed that the HR was 0.53 (95% CI 0.31–0.92) within 30 days and 0.84 (0.60–1.19) after 30 days (appendix 1 p 21).

Medications used during follow-up are summarised in appendix 1 (pp 27–29).

Changes in angina pectoris, assessed by Canadian Cardiovascular Society grades, and in heart function, assessed by New York Heart Association class, between

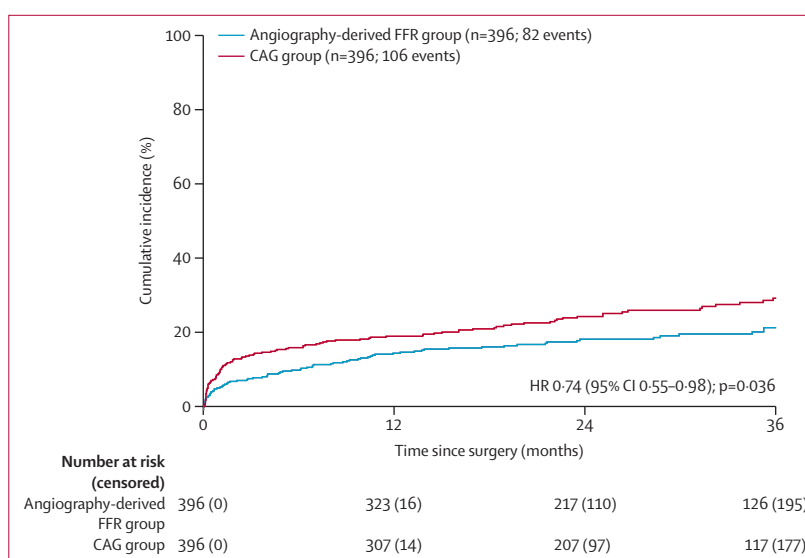


Figure 2: Kaplan-Meier estimates of the key secondary outcome (modified intention-to-treat population)

The key secondary outcome was a composite of all-cause death, myocardial infarction, stroke, unplanned revascularisation, hospitalisation for unstable angina, and hospitalisation for heart failure. There was no significant interaction between randomised treatment and log(time) in the extended Cox model ($p=0.18$). CAG=coronary angiography. FFR=fractional flow reserve.

baseline and 1-year follow-up after surgery are shown in appendix 1 (pp 30–31).

Discussion

In the FAVOR IV-QVAS trial, angiography-derived FFR-guided (physiologically guided) CABG was associated with lower incidence of a composite outcome including death, myocardial infarction, stroke, unplanned revascularisation, and new renal failure requiring dialysis at 30 days, compared with coronary angiography-guided (anatomically guided) CABG. The effect was driven largely by the reduction in postoperative myocardial

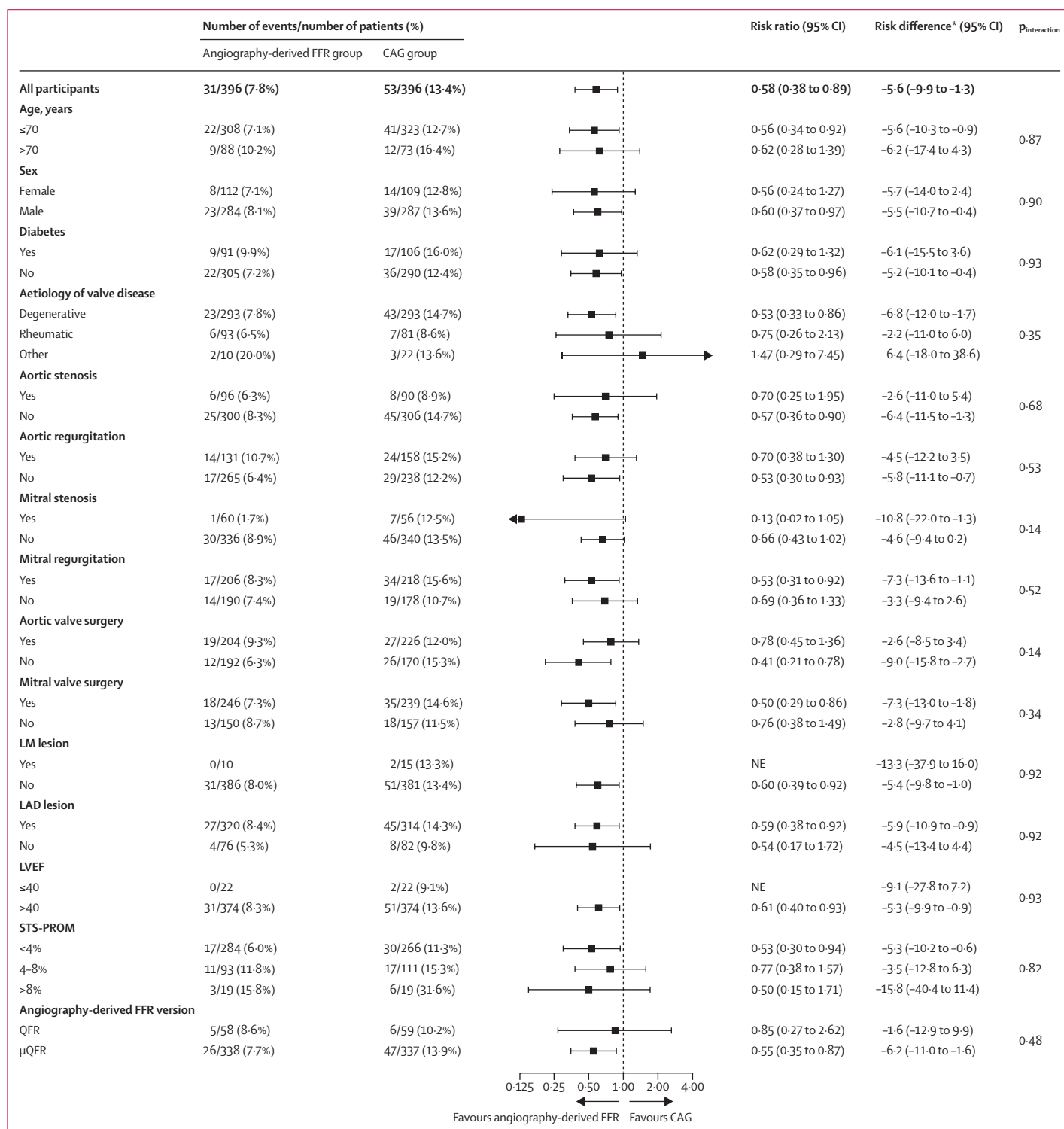


Figure 3: Prespecified subgroup analyses of the primary outcome at 30 days (modified intention-to-treat population)

The primary outcome was a composite of all-cause death, myocardial infarction, stroke, unplanned revascularisation, or new renal failure requiring dialysis within 30 days after surgery. CAG=coronary angiography. FFR=fractional flow reserve. LM=left main coronary artery. LAD=left anterior descending artery. LVEF=left ventricular ejection fraction. NE=not estimable. QFR=quantitative flow ratio. STS-PROM=Society of Thoracic Surgeons predicted risk of mortality. μQFR=upgraded Murray law-based version of QFR. *Risk difference shown as percentage points change.

infarction (appendix 1 p 34) and acute kidney injury. The difference in favour of angiography-derived FFR-guided CABG persisted at a median follow-up of 27 months.

In this trial, almost all patients in the coronary angiography group received concomitant CABG; however, nearly four-fifths of bypassed vessels with visual 50–69% stenosis and one-third of bypassed vessels with visual 70–89% stenosis were physiologically non-significant. Angiography-derived FFR-guided CABG was associated with fewer bypass grafts per patient and with a 15% shorter cardiopulmonary bypass time compared with coronary angiography-guided CABG. Longer duration of systemic non-pulsatile hypoperfusion and cardiac arrest during cardiopulmonary bypass are associated with increasingly higher degrees of ischaemia—reperfusion injury, systemic inflammatory activation, haemolysis, and microembolisation, all of which are thought to contribute causally to postoperative myocardial and renal injury.^{22,23} Postoperative myocardial and renal insults are known to be strongly associated with increased early and late mortality after cardiac surgery.^{24,25} Data from the German Aortic Valve Registry showed that the cardiopulmonary bypass time for surgical aortic valve replacement combined with CABG was, on average, 32 min longer than for isolated aortic valve replacement, and that combined procedures were associated with higher mortality and bleeding rates.²⁶ Similar findings have been reported in the Society of Thoracic Surgeons database.⁶ A possible explanation for these findings is that simplification of the revascularisation strategy, with a concomitant reduction in cardiopulmonary bypass time, reduces intraoperative myocardial and renal injury and thereby improves clinical outcomes. Similar associations between procedural complexity, cardiopulmonary bypass duration, and postoperative outcomes have been observed in randomised trials of mitral valve surgery.²⁷

A potential concern of physiologically guided CABG is the deferral of angiographically severe but physiologically non-flow-limiting lesions, which could then progress and cause subsequent ischaemic events. In a cohort study, CABG was intentionally omitted for angiographically moderate stenoses during primary valve surgery.²⁸ By contrast, in a separate analysis, functional incomplete revascularisation has been associated with increased major adverse cardiovascular events in patients undergoing valve surgery with coronary artery disease.¹² In our trial, the cardiovascular event rates during a median follow-up of 27 months remained lower in the angiography-derived FFR group than in the coronary angiography group, suggesting that the benefits of avoiding unnecessary grafting outweigh the risks of leaving non-ischaemic lesions untreated at least in the first 2 years after surgery. These results are in line with data from percutaneous coronary intervention trials, in which physiologically guided strategies have yielded superior outcomes compared

with anatomy-based approaches. By contrast, no data from randomised clinical trials support physiologically guided CABG in patients with isolated coronary artery disease, although only a few small studies have been published.²⁹

QFR, the method of coronary physiological assessment used in our trial, has recently been incorporated into European guidelines with a class 1B recommendation.³⁰ This approach does not require additional invasive instrumentation and avoids the risks and costs of wire-based physiological measurements. The majority of patients included in our trial were evaluated with the advanced μ QFR technology, which incorporates artificial intelligence and operates from a single angiographic projection.

Our trial has limitations. First, it was conducted exclusively in China, and the reported results might not be generalisable to different patient populations, surgical practices, or health-care systems. Second, the generalisability of our findings is also affected by a substantial proportion of screened patients with ischaemic or other secondary valve disease not being eligible for inclusion and by the relatively low representation of female patients. Third, the absence of formal data and safety monitoring board oversight is a potential limitation. Last, the median duration of follow-up was 27 months, leaving the effect of angiography-derived FFR-guided CABG on longer-term outcomes uncertain; extended follow-up is therefore ongoing.

In conclusion, in patients undergoing valve surgery with concomitant coronary artery disease, physiologically guided CABG using angiography-derived FFR reduced the incidence of the composite perioperative outcome compared with anatomically guided CABG. These differences were maintained during a median follow-up of 27 months, without an increase in adverse cardiovascular events. These findings support a more selective approach to surgical coronary revascularisation guided by physiological assessment in patients undergoing valve surgery.

Contributors

QZ, YZhu, and WZ designed the trial and wrote the protocol with input from SS, BR, and MG. QZ and YZhu obtained the funding and were the trial supervisors. YZhu, ZC, YZha, LH, CZ, SY, LM, CQ, ZG, DZ, XZ, LC, ZW, XY, MZ, HL, JQ, HX, YS, and JZ contributed to data collection. ST provided oversight of the core laboratory analysis. WZ, YZhu, and JZ were responsible for data management. WZ did the statistical analysis. YZhu, WZ, SS, and BR wrote the first draft of the report with input from QZ, MG, ST, ZX, GQ, and WW. YZhu, WZ, and QZ accessed and verified the data. All authors critically revised the manuscript for important intellectual content and vouch for the accuracy and completeness of the data and for the fidelity of the trial to the protocol. All authors had access to all the data in the study and had final responsibility for the decision to submit for publication.

Equitable partnership declaration

The authors of this paper have submitted an equitable partnership declaration (appendix 2). This statement allows researchers to describe how their work engages with researchers, communities, and

See Online for appendix 2

environments in the countries of study. This statement is part of The Lancet Group's broader goal to decolonise global health.

Declaration of interests

QZ reports research grants related to this study from the Shanghai Hospital Development Center and the Ministry of Science and Technology of the People's Republic of China. YZhu reports research grants related to this study from Shanghai Municipal Science and Technology Commission. ST is the cofounder of Pulse Medical, from whom he has received research grants unrelated to this study; he has also been an unpaid consultant for Pulse Medical. All other authors declare no competing interests.

Data sharing

Individual patient data will not be made publicly available but can be accessed on reasonable request to the corresponding author.

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References

- Lappé JM, Grodin JL, Wu Y, Bott-Silverman C, Cho L. Prevalence and prediction of obstructive coronary artery disease in patients referred for valvular heart surgery. *Am J Cardiol* 2015; **116**: 280–85.
- Otto CM, Nishimura RA, Bonow RO, et al. 2020 ACC/AHA Guideline for the management of patients with valvular heart disease: executive summary: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation* 2021; **143**: e35–71.
- Praz F, Borger MA, Lanz J, et al, and the ESC/EACTS Scientific Document Group. 2025 ESC/EACTS Guidelines for the management of valvular heart disease. *Eur J Cardiothorac Surg* 2025; **67**: e2af276.
- Gardner SC, Grunwald GK, Rumsfeld JS, et al. Comparison of short-term mortality risk factors for valve replacement versus coronary artery bypass graft surgery. *Ann Thorac Surg* 2004; **77**: 549–56.
- Beach JM, Mihaljevic T, Svensson LG, et al. Coronary artery disease and outcomes of aortic valve replacement for severe aortic stenosis. *J Am Coll Cardiol* 2013; **61**: 837–48.
- Iribarne A, Zwischenberger B, Hunter Mehaffey J, et al. The Society of Thoracic Surgeons adult cardiac surgery database: 2024 update on national trends and outcomes. *Ann Thorac Surg* 2025; **119**: 1139–50.
- Tonino PA, De Bruyne B, Pijls NH, et al, and the FAME Study Investigators. Fractional flow reserve versus angiography for guiding percutaneous coronary intervention. *N Engl J Med* 2009; **360**: 213–24.
- Xaplanteris P, Fournier S, Pijls NHJ, et al, and the FAME 2 Investigators. Five-year outcomes with PCI guided by fractional flow reserve. *N Engl J Med* 2018; **379**: 250–59.
- Thuesen AL, Riber LP, Veien KT, et al. Fractional flow reserve versus angiographically-guided coronary artery bypass grafting. *J Am Coll Cardiol* 2018; **72**: 2732–43.
- Toth GG, De Bruyne B, Kala P, et al. Graft patency after FFR-guided versus angiography-guided coronary artery bypass grafting: the GRAFFITI trial. *EuroIntervention* 2019; **15**: e999–1005.
- Zhu J, Zhu Y, Zhang W, et al. Preliminary outcomes of quantitative flow ratio-guided coronary bypass grafting in primary valve surgery: a propensity score weighted analysis. *JTCVS Open* 2024; **21**: 90–108.
- Zhang F, Gao W, Wang W, et al. Prognostic value of functional incomplete revascularization in patients undergoing valve surgery with coronary artery disease. *JACC Asia* 2025; **5**: 1273–84.
- Tu S, Westra J, Adedj J, et al. Fractional flow reserve in clinical practice: from wire-based invasive measurement to image-based computation. *Eur Heart J* 2020; **41**: 3271–79.
- Xu B, Tu S, Qiao S, et al. Diagnostic accuracy of angiography-based quantitative flow ratio measurements for online assessment of coronary stenosis. *J Am Coll Cardiol* 2017; **70**: 3077–87.
- Xu B, Tu S, Song L, et al, and the FAVOR III China study group. Angiographic quantitative flow ratio-guided coronary intervention (FAVOR III China): a multicentre, randomised, sham-controlled trial. *Lancet* 2021; **398**: 2149–59.
- Hu X, Zhang J, Yang S, et al, and the FLAVOUR II study group. Angiography-derived fractional flow reserve versus intravascular ultrasound to guide percutaneous coronary intervention in patients with coronary artery disease (FLAVOUR II): a multicentre, randomised, non-inferiority trial. *Lancet* 2025; **405**: 1491–504.
- Andersen BK, Sejr-Hansen M, Maillard L, et al. Quantitative flow ratio versus fractional flow reserve for coronary revascularisation guidance (FAVOR III Europe): a multicentre, randomised, non-inferiority trial. *Lancet* 2024; **404**: 1835–46.
- Fezzi S, Ding D, Scarsini R, et al. Integrated assessment of computational coronary physiology from a single angiographic view in patients undergoing TAVI. *Circ Cardiovasc Interv* 2023; **16**: e013185.
- Tu S, Westra J, Yang J, et al, and the FAVOR Pilot Trial Study Group. Diagnostic accuracy of fast computational approaches to derive fractional flow reserve from diagnostic coronary angiography: the international multicenter FAVOR pilot study. *JACC Cardiovasc Interv* 2016; **9**: 2024–35.
- Tu S, Ding D, Chang Y, Li C, Wijns W, Xu B. Diagnostic accuracy of quantitative flow ratio for assessment of coronary stenosis significance from a single angiographic view: a novel method based on bifurcation fractal law. *Catheter Cardiovasc Interv* 2021; **97** (suppl 2): 1040–47.
- Jacobs JP, Shahian DM, D'Agostino RS, et al. The Society of Thoracic Surgeons National Database 2017 Annual Report. *Ann Thorac Surg* 2017; **104**: 1774–81.
- Paparella D, Cappabianca G, Visicchio G, et al. Cardiac troponin I release after coronary artery bypass grafting operation: effects on operative and midterm survival. *Ann Thorac Surg* 2005; **80**: 1758–64.
- Axtell AL, Fiedler AG, Melnitchouk S, et al. Correlation of cardiopulmonary bypass duration with acute renal failure after cardiac surgery. *J Thorac Cardiovasc Surg* 2020; **159**: 170–178.e2.
- Gaudino M, Flather M, Capodanno D, et al. European Association of Cardio-Thoracic Surgery (EACTS) expert consensus statement on perioperative myocardial infarction after cardiac surgery. *Eur J Cardiothorac Surg* 2024; **65**: ezad415.
- Cheruku SR, Raphael J, Neyra JA, Fox AA. Acute kidney injury after cardiac surgery: prediction, prevention, and management. *Anesthesiology* 2023; **139**: 880–98.
- Fujita B, Ensminger S, Bauer T, et al, and the GARY Executive Board. Trends in practice and outcomes from 2011 to 2015 for surgical aortic valve replacement: an update from the German Aortic Valve Registry on 42 776 patients. *Eur J Cardiothorac Surg* 2018; **53**: 552–59.
- Acker MA, Parides MK, Perrault LP, et al, and the CTSN. Mitral-valve repair versus replacement for severe ischemic mitral regurgitation. *N Engl J Med* 2014; **370**: 23–32.
- Del Forno B, Ascione G, Lapenna E, et al. Is myocardial revascularization really necessary in patients with $\geq 50\%$ but $< 70\%$ coronary stenosis undergoing valvular surgery? *Eur J Cardiothorac Surg* 2020; **58**: 343–49.
- Spadaccio C, Glineur D, Barbato E, et al. Fractional flow reserve-based coronary artery bypass surgery: current evidence and future directions. *JACC Cardiovasc Interv* 2020; **13**: 1086–96.
- Vrints C, Andreotti F, Koskinas KC, et al, and the ESC Scientific Document Group. 2024 ESC Guidelines for the management of chronic coronary syndromes. *Eur Heart J* 2024; **45**: 3415–537.